LECTURE NOTES ON ORDINARY DIFFERENTIAL EQUATIONS

ALEJANDRO KOCSARD

1. Bibliography

The main reference for this course is the book of Viana and Espinar [VE21]. Complementary references are the books of Sotomayor [Sot79] and Arnold [Arn06].

2. Introduction

A **differentiable equation** is an equation whose unknown is a differentiable function and that relates some of their derivatives. When the unknown of the equation is a function of a real variable, we say its is an **ordinary differential equation**, or **ODE** for short. On the other hand, when the unknown is a differentiable function of several real variables and the equation established a relation among its partial derivatives, one says it is a **partial differential equation**, or **PDE** for short. In this course we shall focus on ordinary differential equations.

More precisely, an ODE is an expression

$$F\big(t, x, x^{(1)}, \dots, x^{(k)}\big) = 0, \tag{1}$$

where $F\colon \mathcal{U} \subset \mathbb{R}^{1+(k+1)d} \to \mathbb{R}^d$ is a continuous function, $\mathcal{U}$ is an open subset, $t \in \mathbb{R}$ and $x, x^{(1)}, \dots x^{(k)} \in \mathbb{R}^d$. A **solution** of (1) is a C^k function $x\colon I \to \mathbb{R}^d$ such that I is a non-empty open interval and satisfying

$$\big(t, x(t), x'(t), \dots, x^{(k)}(t)\big) \in \mathcal{U}, \quad \forall t \in I,$$

and

$$F\big(t, x(t), \dots, x^{(k)}(t)\big) = 0, \quad \forall t \in I.$$

All along these notes, we shall the following notations for the kth-order derivative of (the C^k function) $x(t)$:

$$x^{(k)}(t) = \frac{\mathrm{d}^k x}{\mathrm{d}t^k}(t).$$

For $k = 1, 2$ we shall also use the classical notation

$$x'(t) = \dot{x}(t) = x^{(1)}(t) = \frac{\mathrm{d}x}{\mathrm{d}t}(t),$$

and

$$x''(t) = \ddot{x}(t) = x^{(2)}(t) = \frac{\mathrm{d}^2 x}{\mathrm{d}^2 t}(t).$$

Example 2.1. Consider the function $F\colon \mathbb{R}^3 \to \mathbb{R}^1$ given by

$$F(t, x, x^{(1)}) := x^{(1)} - 2x, \quad \forall (t, x, x^{(1)}) \in \mathbb{R}^3.$$

So, the ODE $F \equiv 0$ has infinitely many solutions. In fact, for each $C \in \mathbb{R}$ the function $x(t) := Ce^{2t}$, for every $t \in \mathbb{R}$, is a solution and in fact, these are all solutions.

Date: June 22, 2025.

Example 2.2. The following ODE

$$\big(x'\big)^2 + \big(x\big)^2 + 1 = 0$$

has clearly no solution.

Example 2.3. The following ODE

$$\big(x''\big)^2 + (x-1) = 0,$$

has only one solution, the constant function $x \equiv 1$.

So, this ODEs are maybe too general, and thus, we shall restrict ourselves to ODEs written in *explicit form:* from now on, an *ODE of order* $k \in \mathbb{N}$ *in* $\mathbb{R}^d$ is an expression

$$x^{(k)} = f\big(t, x, x^{(1)}, \dots, x^{(k-1)}\big), \tag{2}$$

where $f\colon U \subset \mathbb{R}^{1+kd} \to \mathbb{R}^d$ is a continuous function and $U \subset \mathbb{R}^{1+dk}$ is an open subset. A solution of this ODE is C^k function $x\colon I \to \mathbb{R}^d$, where I is an open interval, $\big(t, x(t), x'(t), \dots, x^{(k-1)}(t)\big) \in U$, for all $t \in I$ and

$$x^{(k)}(t) = f\big(t, x(t), x'(t), \dots, x^{(k-1)}(t)\big), \quad \forall t \in I.$$

3. Solving and studying ODEs

So, in this course we shall study *ordinary differential equations,* ODEs for short. But, what does it mean to *study or to solve* and ODEs?

Classically, for Newton and Leibniz in the XVIIth century the study of an ODE consisted in finding explicit formulas for all its solutions. In modern language, for them *to solve an ODE* meant to solve it **symbolically** by the so called *quadrature methods*. A prototypical example of a quadrature method is the *separation of variables:*

Example 3.1. Consider the differential equation

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x(x-1).$$

Invoking the so called *separation of variables method* one gets

$$\frac{\mathrm{d}x}{x(x-1)} = \mathrm{d}t,$$

and so for every constant $c \in \mathbb{R}$ we get a solution $x(t)$ satisfying

$$\log\left|\frac{x(t)-1}{x(t)}\right| = t + c.$$

Hence, the possible solutions are

$$x(t) := \begin{cases} \big(1 - e^{t+c}\big)^{-1}, & \text{for } t \in (-\infty, -c); \\ \big(1 - e^{t+c}\big)^{-1}, & \text{for } t \in (-c, +\infty); \\ \big(1 + e^{t+c}\big)^{-1}, & \text{for } t \in \mathbb{R}. \end{cases}$$

We shall revisit Example 3.1 in several occasions.

Another remarkable problem that can be symbolically solved in the celebrated

Example 3.2 (Restricted two-body problem)**.** The *"restricted two-body problem"* refers to a simplified model used to analyze the motion of two celestial bodies under the influence of their mutual gravitational attraction, where one body is significantly more massive than the other. In this framework, the more massive body is assumed to be stationary, while the second body is treated as a point mass that moves in the gravitational field created by the first one. This approach allows for the

derivation of equations of motion for the second body without needing to account for its gravitational effects on the primary, simplifying the analysis. This model is particularly useful in celestial mechanics, where it can be applied to scenarios such as a satellite orbiting a planet or a comet passing by a star.

More precisely, the *restricted two-body problem* consider in describing the orbit of massive point p, whose location in the space at time $t \in \mathbb{R}$ is given by $x(t) \in \mathbb{R}^3$, moving in a central field force in $\mathbb{R}^3$ that is generated by a massive point of mass $M > 0$ which is fixed at the origin. According to Newton's laws, the trajectory $x(t) \in \mathbb{R}^3$ is the solution of the following initial value problem:

$$\begin{cases} \ddot{x}(t) = -\dfrac{GM}{\|x\|^3} x(t), \\ x(0) = x_0, \\ \dot{x}(0) = v_0, \end{cases} \tag{3}$$

where $G > 0$ is the universal constant of gravitation, $x_0 \in \mathbb{R}^3$ and $v_0 \in \mathbb{R}^3$ are the initial position and velocity, respectively.

Using classical *quadrature techniques* one can show that the solutions $x(t)$ of (3) are either straight lines through the origin (when x_0 and v_0 are collinear vectors in $\mathbb{R}^3$), or quadratic curves (i.e. ellipses, parabolas or hyperbolas).

However, short after the pioneering works of Newton and Leibniz, it became pretty clear that was impossible to solve symbolically every ODE, even though most of the explicit ODEs given by *elementary functions*. A classical example is given by the well-known *Gauss' bell function:*

Example 3.3. No *elementary function* $x\colon \mathbb{R} \to \mathbb{R}$ is a solution for the ODE

$$x'(t) = e^{-t^2}.$$

So, another strategy was needed in order to *study ODEs*. And it is at that time that **numerical methods** started to be developed: if we cannot explicitly write the solution of certain ODE, at least we can get *numerical approximations:*

Problem 3.1. Use the so called *Euler method* to get numerical approximations of $x(1)$ where the function $x\colon \mathbb{R} \to \mathbb{R}$ is given by

$$x(t) := \int_0^t e^{-s^2}\, \mathrm{d}s, \quad \forall t \in \mathbb{R}.$$

However, the numerical methods have a very high computational cost when *"the domain is very large"* and hence, for instance in Problem 3.1, it more and more expensive to compute numerically $x(t)$, as $t \to +\infty$. On the other hand, there many problems that are more *qualitative than quantitative* and, in general, numerical methods (and even the symbolic method) are not appropriate for this kind of questions.

The following example is probably the main original motivation for the development of the so called **geometric method,** mainly due to Poincaré:

Example 3.4 (n-body problem)**.** The *n-body problem* refers to the challenge of predicting the individual motions of a group of n celestial objects that interact with each other through gravitational forces. The complexity arises from the fact that each body exerts a gravitational force on every other body, leading to a system of coupled differential equations. If $x_i(t) \in \mathbb{R}^3$ represents the position of the i-th object at time $t \in \mathbb{R}$ and $m_i > 0$ denotes its mass, then Newton's second law determines that these quantities are related by the following system of ordinary differential

equations

$$\begin{cases} \ddot{x}_i(t) = \sum_{j\neq i} G \frac{m_j}{\|x_i(t)-x_j(t)\|^3}\big(x_i(t)-x_j(t)\big), \\ x_i(0) = x_{i,0}, \\ \dot{x}_i(0) = v_{i,0}, \end{cases} \tag{4}$$

for $1 \leqslant i \leqslant n$, and where $x_{i,0}, v_{i,0} \in \mathbb{R}^3$ denotes the position and the velocity of the i-th object at time $t = 0$, respectively.

It is pretty remarkable that, in general, the system of ODEs given by (4) cannot be solved by "quadrature methods" when $n \geqslant 3$. In fact, their solutions are not elementary functions in general.

A classical and famous *qualitative problem* about Example 3.4 consists in determining whether the trajectory of a certain object (with certain initial conditions) is **periodic** or not. For instance, is the orbit of Earth really a closed curve in space?

Another important ODE that cannot solved by *quadrature methods* and has important qualitative questions associated to it, is the so called *harmonic pendulum* (see [VE21, Example 1.3]).

On the other hand, in certain cases, the classical symbolic methods can exhibit some limitations and it could be rather complicated to extract interesting information from the solutions that they produce. For instance, in Example 3.1 the constant solutions $x \equiv 0$ and $x \equiv 1$ cannot be gotten by the separation of variables. Moreover, it is rather complicated to understand the qualitative behavior of all solutions just regarding their formulas. See [VE21, Example 1.4] for more details and for a qualitative description of solutions.

4. Exercises

Some important exercises from [VE21, Chapter 1]:
1.1, 1.2, 1.3, 1.10, 1.11, 1.12, 1.13, 1.15, 1.17.

5. Existence and uniqueness of solutions

In Example 2.2 we saw that some ODEs have no solution at all. However, will see that under very mild conditions every ODE in explicit form admits a solution and in fact, at a certain instance, the solutions are unique.

In order to study the existence and uniqueness of solutions for ODEs written in explicit form, it is important to notice first that any ODE of order k can be rewritten as a first order ODE. In fact, let us suppose that $f\colon U \subset \mathbb{R}^{1+dk} \to \mathbb{R}^d$ is a continuous function and consider the ODE given by (2). Then, we can consider the function $\tilde{f}\colon U \subset \mathbb{R}^{1+dk} \to \mathbb{R}^{dk}$ give by

$$\tilde{f}(t, x, x^{(1)}, x^{(2)}, \dots, x^{(k-1)}) = \big(x^{(2)}, \dots, x^{(k-2)}, f(t, x^{(1)}, \dots, x^{(k-1)})\big), \tag{5}$$

for all $(t, x^{(1)}, \dots, x^{(k-1)}) \in U$, and one can easily check that $x\colon I \to \mathbb{R}^d$ is a solution of equation (2) if and only if the function $X\colon I \to \mathbb{R}^{kd}$ given by

$$X(t) := \big(x(t), x'(t), \dots, x^{k-1}(t)\big), \quad \forall t \in I,$$

is a solution of the first order ODE

$$X' = \tilde{f}(t, X).$$

5.1. **Picard theorem on existence and uniqueness of solutions.** Let $U \subset \mathbb{R}^{1+d}$ be a non-empty open set and $f\colon U \to \mathbb{R}^d$ be a continuous function. We say that the function f is **locally Lipschitz on the second variable** when the following holds: for every compact subset $K \subset U$, there is real constant $L_K > 0$ such that

$$\|f(t,x_2) - f(t,x_1)\| \leqslant L_K \|x_2 - x_1\|, \quad \forall (t,x_1),\ (t,x_2) \in K \subset U. \tag{6}$$

On the other hand, given a point $(t_0,x_0) \in U$ and real numbers $a,b > 0$, we shall say that $B_a(t_0) \times B_b(x_0)$ is a **good box centered at** (t_0,x_0) when

$$\overline{B_a(t_0)} \times \overline{B_b(x_0)} \subset U,$$

and

$$\frac{b}{a} \geqslant \max\left\{\|f(t,x)\| : (t,x) \in \overline{B_a(t_0)} \times \overline{B_b(x_0)}\right\}.$$

For simplicity, we shall just say that $B_a(t_0) \times B_b(x_0)$ is a **good box centered at** (t_0,x_0) when is a good M-box, for some $M > 0$.

Lemma 5.1 (Existence of good boxes)**.** *Given any continuous function $f\colon U \subset \mathbb{R}^{1+d} \to \mathbb{R}^d$ as above, for every $(t_0,x_0) \in U$ there is a good box centered at (t_0,x_0).*

Proof. It is left as an exercise to the reader. □

Theorem 5.2 (Picard's theorem)**.** *If $U \subset \mathbb{R}^{1+d}$ is an open set and $f\colon U \to \mathbb{R}^d$ is a continuous and locally Lipschitz on the second variable function, then for every $(t_0,x_0) \in U$ and any good box $B_a(t_0) \times B_b(x_0)$ centered at (t_0,x_0), there is a unique C^1 function $\gamma\colon B_a(t_0) \to \mathbb{R}^d$ which is a solution of the* ***initial value problem*** IVP (f,t_0,x_0)

$$\begin{cases} x'(t) = f\big(t,x(t)\big), \\ x(t_0) = x_0. \end{cases} \tag{7}$$

Proof of Theorem 5.2. Let us consider the functional space

$$\mathscr{C} := \left\{\phi \in C^0\big(\overline{B_a(t_0)}, \overline{B_b(x_0)}\big) : \phi(t_0) = x_0\right\},$$

which is endowed with the distance function $d\colon \mathscr{C} \times \mathscr{C} \to \mathbb{R}$ given by

$$d(\phi_1,\phi_2) := \max\left\{\|\phi_1(t) - \phi_2(t)\| : t \in \overline{B_a(t_0)}\right\}, \quad \forall \phi_1,\phi_2 \in \mathscr{C}.$$

Then, given any $\phi \in \mathscr{C}$, consider the new function $\mathcal{L}\phi\colon \overline{B_a(t_0)} \to \mathbb{R}^d$ given by

$$\mathcal{L}\phi(t) := x_0 + \int_{t_0}^{t} f\big(s,\phi(s)\big)\,\mathrm{d}s, \quad \forall s \in \overline{B_a(t_0)}.$$

Exercise 5.1. Show that $(\mathscr{C},d)$ is a complete metric space. On the other hand, $\mathcal{L}\phi \in \mathscr{C}$, for every $\phi \in \mathscr{C}$ and $\gamma \in \mathscr{C}$ is a solution of the initial value problem IVP(f,t_0,x_0) given by (7) if and only if $\mathcal{L}\gamma = \gamma$.

Then, we are going to prove that there exist $k \in \mathbb{N}$ and $\lambda \in (0,1)$ such that the iterated operator $\mathcal{L}^k := \mathcal{L}^{k-1} \circ \mathcal{L}$ satisfies

$$d\big(\mathcal{L}^k\phi_1, \mathcal{L}^k\phi_2\big) \leqslant \lambda d(\phi_1,\phi_2), \quad \forall \phi_1,\phi_2 \in \mathcal{C}. \tag{8}$$

Exercise 5.2. Show that condition (8) implies that $\mathcal{L}$ has a exactly one fixed point, i.e. there is a unique $\gamma \in \mathcal{C}$ such that $\mathcal{L}\gamma = \gamma$.

So, in order to finish the proof we just need to prove estimate (8). To do that, observe that if $L > 0$ denotes the Lipschitz constant of f for the second coordinate on the compact set $\overline{B_a(t_0)} \times \overline{B_b(x_0)}$ given by condition (6), the it hods

$$\begin{aligned}\left\|\mathcal{L}^k\phi_1(t) - \mathcal{L}^k\phi_2(t)\right\| &\leqslant \int_{t_0}^{t} \left\|f\big(s_1, \mathcal{L}^{k-1}\phi_1(s_1)\big) - f\big(s_1, \mathcal{L}^{k-1}\phi_2(s_1)\big)\right\| \,\mathrm{d}s_1 \\ &\leqslant L\int_{t_0}^{t} \left\|\mathcal{L}^{k-1}\phi_1(s_1) - \mathcal{L}^{k-1}\phi_2(s_1)\right\| \,\mathrm{d}s_1 \\ &\leqslant L^2\int_{t_0}^{t}\int_{t_0}^{s_1} \left\|\mathcal{L}^{k-2}\phi_1(s_2) - \mathcal{L}^{k-2}\phi_2(s_2)\right\| \,\mathrm{d}s_2,\end{aligned}$$

and by induction we get

$$\begin{aligned}\left\|\mathcal{L}^k\phi_1(t) - \mathcal{L}^k\phi_2(t)\right\| &\leqslant L^k\int_{t_0}^{t}\int_{t_0}^{s_1}\int_{t_0}^{s_2}\cdots\int_{t_0}^{s_{k-1}} \left\|\phi_1(s_k) - \phi_2(s_k)\right\| \,\mathrm{d}s_k \\ &\leqslant L^k d(\phi_1,\phi_2)\frac{|t-t_0|^k}{k!} \leqslant \lambda d(\phi_1,\phi_2),\end{aligned}$$

where

$$\lambda := \frac{L^k a^k}{k!} < 1,$$

provided $k \in \mathbb{N}$ is sufficiently large. □

Corollary 5.3. *If f and (t_0, x_0) are as in Theorem 5.2 and $\gamma_i \colon I_i \to \mathbb{R}^d$ are solution of the* IVP(f, t_0, x_0) *given by (7) for $i = 1, 2$, then it holds*

$$\gamma_1(t) = \gamma_2(t), \quad \forall t \in I_1 \cap I_2.$$

Proof. Let us define $I = \{t \in I_1 \cap I_2 : \gamma_1(t) = \gamma_2(t)\}$. The set I is clearly non-empty because $t_0 \in I$ and since γ_1 and γ_2 are continuous, I is closed in $I_1 \cap I_2$. On the other hand, by Theorem 5.2 I is also open in $I_1 \cap I_2$. Since the intersection of two intervals in always connected, we conclude that $I = I_1 \cap I_2$, as desired. □

6. Existence of solutions

In this section we shall prove so called Peano's theorems that claims the existence of solutions for initial value problems under weaker hypotheses than those of Picard's theorems (Theorem 5.2). But before doing that, let us study the following

Example 6.1. Let $f \colon \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(t, x) := \sqrt[3]{x^2}, \quad \forall (t, x) \in \mathbb{R}^2.$$

One can easily check that $\gamma(t) := 0$ and $\phi(t) := t^3/3$ for every $t \in \mathbb{R}$ are two (different) solutions for the initial value problem IVP$(f, 0, 0)$.

Observe that this function f is continuous, but it is not locally Lipschitz on the second variable at $(t, 0)$, for every $t \in \mathbb{R}$. So, this example does not contradict Theorem 5.2.

Even though we cannot guarantee uniqueness of solutions for continuous functions, we have the following

Theorem 6.2 (Peano's theorem)**.** *Let $U \subset \mathbb{R}^{1+d}$ be a open set and $f \colon U \to \mathbb{R}^d$ be a continuous function. Then, for every $(t_0, x_0) \in U$ and any good box $B_a(t_0) \times B_b(x_0)$ centered at (t_0, x_0), there is a solution $\gamma \colon B_a(t_0) \to \mathbb{R}^d$ for the initial value problem* IVP(f, t_0, x_0).

See [VE21, §2.2] for a proof. Here we can sketch a different one:

Proof of Theorem 6.2. This proof is essentially motivated by so called *Euler's method.* For any $n \in \mathbb{N}$ and each $i \in \{-n, -n+1, \dots, 0, 1, \dots, n\}$ consider the points

$$t_{i,n} := t_0 + \frac{ia}{n} \in \overline{B_a(t_0)},$$

and inductively defined $x_{0,n} := x_0$, and for $i \geqslant 1$,

$$x_{i,n} := x_0 + \frac{a}{n} \sum_{j=0}^{i-1} f(t_{j,n}, x_{j,n}),$$

and for $i \leqslant -1$,

$$x_{i,n} := x_0 - \frac{a}{n} \sum_{j=0}^{|i|-1} f(t_{-j,n}, x_{-j,n}),$$

Then, for each $n \in \mathbb{N}$ we consider the function $\gamma_n \colon \overline{B_a(t_0)} \to \overline{B_b(x_0)} \subset \mathbb{R}^d$ given by

$$\gamma_n(t) := x_{i,n} + (t - t_{i,n}) f(t_{i,n}, x_{i,n}), \quad \forall t \in [t_{i,n}, t_{i+1,n}],$$

and each $i \in \{-n, \dots, n-1\}$.

Thus, we can apply Arzelà-Ascoli theorem to guarantee there is subsequence $(\gamma_{n_j})_{j \geqslant 1}$ and a continuous function $\gamma \colon \overline{B_a(t_0)} \to \overline{B_b(x_0)}$ such that $\gamma_{n_j} \to \gamma$, uniformly on $\overline{B_a(t_0)}$, as $n_j \to +\infty$. □

7. Maximal solutions

We can apply Corollary 5.3 to define *maximal solutions.* In fact, a solution $\gamma \colon I \to \mathbb{R}^d$ of the initial value problem $\mathrm{IVP}(f, t_0, x_0)$ given by (7) is said to be a **maximal solution** when I is an open interval, γ is a solution of $\mathrm{IVP}(f, t_0, x_0)$ and given any other solution $\phi \colon J \to \mathbb{R}^d$ of $\mathrm{IVP}(f, t_0, x_0)$, it holds $J \subset I$ and $\gamma(t) = \phi(t)$, for all $t \in J$.

Corollary 7.1 (Existence of maximal solutions)**.** *If f and (t_0, x_0) are as in Theorem 5.2, there exists a unique maximal solution $\gamma_{t_0,x_0} \colon I_{t_0,x_0} \to \mathbb{R}^d$ of the initial value problem* $\mathrm{IVP}(f, t_0, x_0)$ *given by* (7)*.*

Proof. Let $S := \{\gamma_\alpha \colon I_\alpha \to \mathbb{R}^d : \gamma_\alpha \text{ is a solution of } \mathrm{IVP}(f, t_0, x_0)\}_{\alpha \in \Lambda}$ be the set of all solutions of $\mathrm{IVP}(f, t_0, x_0)$. Then we define

$$I_{t_0,x_0} := \bigcup_{\alpha \in \Lambda} I_\alpha,$$

and $\gamma_{t_0,x_0} \colon I_{t_0,x_0} \to \mathbb{R}^d$ by

$$\gamma_{t_0,x_0}(t) := \gamma_\alpha(t), \quad \forall t \in I_\alpha, \ \forall \alpha \in \Lambda.$$

Observe that by Corollary 5.3, function γ_{t_0,x_0} is well-defined and it clearly the maximal solution. □

We can combine Picard's theorem (Theorem 5.2) and Corollary 7.1 to prove that every maximal solution converges to the boundary of the domain U:

Theorem 7.2 (Escaping from compact sets)**.** *Let U and f be as in Theorem 5.2, (t_0, x_0) an arbitrary point of U and $\gamma_{t_0,x_0} \colon I_{t_0,x_0} \to \mathbb{R}^d$ the corresponding maximal solution of the initial value problem* $\mathrm{IVP}(f, t_0, x_0)$*. If $I_{t_0,x_0} = (a, b)$ with $a, b \in \mathbb{R}$, then for every non-empty compact subset $K \subset U$, there is a $\varepsilon > 0$ such that*

$$\big(t, \gamma_{t_0,x_0}(t)\big) \notin K, \quad \forall t \in (a, a+\varepsilon) \cup (b-\varepsilon, b).$$

An analogous result holds for $I_{t_0,x_0} = (-\infty, b)$ or $I_{t_0,x_0} = (a, +\infty)$

Proof of Theorem 7.2. Let us start considering the following

Exercise 7.1. Let $f\colon U \subset \mathbb{R}^{1+d} \to \mathbb{R}^d$ be a continuous function, where U is open, and $K \subset U$ be a compact subset. Then, there is a real numbers $r_K, h_K > 0$ such that

$$B_{r_K}(t) \times B_{h_K}(x)$$

is a good box for f, for every $(t,x) \in K$.

Now, reasoning by contradiction, let us suppose there exists a compact subset $K \subset U$ and a real number $t_1 \in I_{t_0,x_0}$ such that $t_1 \in (b, b - r_K)$, where r_k is the positive real number given by Exercise 7.1 and $(t_1, x_1) \in K$, where $x_1 := \gamma_{t_0,x_0}(t_1)$. Then, by Exercise 7.1 $B_{r_k}(t_1) \times B_{h_k}(x_1)$ is a good box centered at the point (t_1, x_1) and by Theorem 5.2, we know that $\mathrm{IVP}(f, t_1, x_1)$ has a unique solution defined on the set $B_{r_k}(t_1)$. Since $t_1 + r_k > b$ and $x_1 = \gamma_{t_0,x_0}(t_1)$, we conclude that b belongs to the domain of the maximal solution of $\mathrm{IVP}(f, t_0, x_0)$, that we are denoting by I_{t_0,x_0}, contradicting the fact that I_{t_0,x_0} were equal to (a, b). The other cases can be proven in a completely analogous way. □

Definition 1 (Complete equations)**.** Given function $f\colon U \subset \mathbb{R}^{1+d} \to \mathbb{R}^d$ satisfying the hypotheses of Theorem 5.2, we say that the corresponding ODE $x' = f(t,x)$ is **complete** when it holds $I_{t_0,x_0} = \mathbb{R}$, for every $(t_0, x_0) \in U$.

8. Theorems on continuous dependence

In this section we shall consider **parameterized families of ODEs.** More precisely, let $U \subset \mathbb{R}^{1+d+p}$ be an open subset and $f\colon U \to \mathbb{R}^d$ be a continuous function that is **locally Lipschitz on its second variable,** i.e. for every compact subset $K \subset U$, there is a positive constant $L_K > 0$ such that

$$\|f(t,\bar{x},\mu) - f(t,x,\mu)\| \leqslant L_K \|\bar{x} - x\|, \quad \forall (t,\bar{x},\mu),(t,x,\mu) \in K.$$

Given any $(t_0, x_0, \mu_0) \in U$, we shall write $\gamma_{t_0,x_0,\mu_0}\colon I_{t_0,x_0,\mu_0} \to \mathbb{R}^d$ for the maximal solution of the initial value problem $\mathrm{IVP}\big(f(\cdot,\cdot,\mu_0), t_0, x_0\big)$ given by

$$\begin{cases} x'(t) = f\big(t, x(t), \mu_0\big), \\ x(t_0) = x_0. \end{cases}$$

Given constants $a, b, c > 0$ and any point $(t_0, x_0, \mu_0) \in U$, we say that $B_a(t_0) \times B_b(x_0) \times B_c(\mu_0)$ is a **good parametric box** centered at (t_0, x_0, μ_0) when

$$\overline{B_a(t_0)} \times \overline{B_b(x_0)} \times \overline{B_c(\mu_0)} \subset U,$$

and

$$\frac{b}{a} \geqslant \max\left\{\|f(t,x,\mu)\| : (t,x,\mu) \in \overline{B_a(t_0)} \times \overline{B_b(x_0)} \times \overline{B_c(\mu_0)}\right\}.$$

Exercise 8.1 (Existence of parametric good boxes)**.** Given f as above, show that given an arbitrary point $(t_0, x_0, \mu_0) \in U$, there is a good parametric box centered at (t_0, x_0, μ_0).

As a straight forward consequence of Picard's theorem (Theorem 5.2) and Exercise 8.1, we conclude that given any $(t_0, x_0, \mu_0) \in U$, there is $a > 0$ such that $(t_0, x_0, \mu) \in U$ and

$$\overline{B_a(t_0)} = [t_0 - a, t_0 + a] \subset I_{t_0,x_0,\mu}, \quad \forall \mu \in B_a(\mu_0).$$

Theorem 8.1 (Local theorem of continuous dependence on parameters)**.** *If f, $(t_0, x_0, \mu_0) \in U$ and $a > 0$ are as above, then for every $\bar{\mu} \in B_a(\mu_0)$, it holds*

$$\gamma_{t_0,x_0,\mu}(t) \to \gamma_{t_0,x_0,\bar{\mu}}(t), \quad \forall t \in \overline{B_a(t_0)},$$

as $\mu \to \bar{\mu}$.

Proof. Let $a, b, c > 0$ be real constants such that $B_a(t_0) \times B_b(x_0) \times B_c(\mu_0)$ is a parametric good box centered at (t_0, x_0, μ_0), and let us write

$$M := \max\left\{\|f(t,x,\mu)\| : (t,x,\mu) \in \overline{B_a(t_0)} \times \overline{B_b(x_0)} \times \overline{B_c(\mu_0)}\right\}.$$

Let $\bar{\mu}$ be an arbitrary point of $B_c(\mu)$ and consider a sequence $(\mu_n)_{n\geqslant 1} \subset B_c(\mu_0)$ such that $\mu_n \to \bar{\mu}$, as $n \to +\infty$.

Since $B_a(t_0) \times B_b(x_0) \times B_c(\mu_0)$ is a good parametric box, it holds $\gamma_{t_0,x_0,\mu}(t) \in \overline{B_b(x_0)}$ for every $\mu \in \overline{B_c(\mu_0)}$ and every $t \in \overline{B_a(t_0)}$. In particular this implies that $\left|\gamma'_{t_0,x_0,\mu}(t)\right| \leqslant M$ and $\gamma_{t_0,x_0,\mu}(t_0) = x_0$, for every $t \in B_a(t_0)$ and any $\mu \in \overline{B_c(\mu_0)}$. So, we can apply Arzelà-Ascoli theorem to conclude there is a continuous function $\phi\colon \overline{B_a(t_0)} \to \mathbb{R}^d$ and a subsequence (μ_{n_j}) such that $\gamma_{t_0,x_0,\mu_{n_j}}(t) \to \phi(t)$, uniformly on $t \in \overline{B_a(t_0)}$, as $n_j \to +\infty$. Thus, by Leibniz theorem we have that

$$\begin{aligned}\phi(t) &= \lim_{n_j\to+\infty} \gamma_{t_0,x_0,\mu_{n_j}}(t) = \lim_{n_j\to+\infty}\left(x_0 + \int_{t_0}^t f\big(s, \gamma_{t_0,x_0,\mu_{n_j}}(s), \mu_{n_j}\big)\,\mathrm{d}s\right)\\ &= x_0 + \int_{t_0}^t f\big(s, \phi(s), \bar{\mu}\big)\,\mathrm{d}s.\end{aligned}$$

So, ϕ is a fixed point of Picard's operator and hence, $\phi = \gamma_{t_0,x_0,\bar{\mu}}$ on $B_a(t_0)$. Since every sequence (μ_n) contains a subsequence that converges to the same limit, we conclude the original sequence converges to the limit $\gamma_{t_0,x_0,\bar{\mu}}$. □

In order to prove a global version of continuous dependence on parameters, we first need the following

Exercise 8.2. Let $f\colon U \subset \mathbb{R}^{1+d+p} \to \mathbb{R}^d$ be a continuous function, where U is an open set. Let $(t_0, x_0, \mu_0) \in U$ be an arbitrary point and $\gamma\colon I \to \mathbb{R}^d$ any solution of the initial value problem $\mathrm{IVP}\big(f(\cdot,\cdot,\mu_0), t_0, x_0\big)$. Then, given any $t \in I$, there are sequences $t_0, t_1, \ldots, t_n = t$, $a_0, \ldots, a_n$, $b_0, \ldots, b_n$ and $c_0, \ldots, c_n$ such that $t_i \in I$, $a_i, b_i, c_i > 0$ and $B_{a_i}(t_i) \times B_{b_i}\big(\gamma(t_i)\big) \times B_{c_i}(\mu_0)$ is a good parametric box, for every $i \in \{0, 1, \ldots, n\}$ and $\big(t_i, \gamma(t_i)\big) \in B_{a_{i-1}}(t_{i-1}) \times B_{b_{i-1}}\big(\gamma(t_{i-1})\big)$, for every $i \in \{1, \ldots, n\}$.

Now we can prove the following global version of Theorem 8.1:

Theorem 8.2 (Global theorem of continuous dependence on parameters)**.** *Let f, (t_0, x_0, μ_0) be as in Theorem 8.1. Then, for every $t \in I_{t_0,x_0,\mu_0}$, where $\gamma_{t_0,x_0,\mu_0}\colon I_{t_0,x_0,\mu_0} \to \mathbb{R}^d$ denotes the maximal solution, there is a real number $c > 0$ such that $t \in I_{t_0,x_0,\bar{\mu}}$, for every $\bar{\mu} \in B_c(\mu_0)$ and it holds*

$$\gamma_{t_0,x_0,\mu}(t) \to \gamma_{t_0,x_0,\bar{\mu}}(t), \quad \text{as } \mu \to \bar{\mu}.$$

Proof. This a straight forward consequence of Theorem 8.1 and Exercise 8.2. □

As a consequence of Theorem 8.2 we get the continuous dependence on initial conditions:

Theorem 8.3 (Continuous dependence on initial conditions)**.** *Let $f\colon U \subset \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}^d$ be a function like in Theorem 5.2. For each $(t_0, x_0) \in U$, let $\gamma_{t_0,x_0}\colon I_{t_0,x_0} \to \mathbb{R}^d$ be the maximal solution for the initial value problem* $\mathrm{IVP}(f, t_0, x_0)$*. Then, the* global domain

$$\mathcal{D}(f) := \left\{(t_0, x_0, t) \in \mathbb{R}^{1+d+1} : (t_0, x_0) \in U,\ t \in I_{t_0,x_0}\right\}$$

is an open subset of $\mathbb{R}^{1+d+1}$ and the global solution *$\phi\colon \mathcal{D}(f) \to \mathbb{R}^d$ given by*

$$\phi(t_0, x_0, t) := \gamma_{t_0,x_0}(t), \quad \forall (t_0, x_0, t) \in \mathcal{D}(f),$$

is a continuous function.

Proof. In order to prove that the global solution ϕ is continuous, let us consider the following parametric family of ODEs given by $F\colon V\colon \mathbb{R}^{1+d+(1+d)} \to \mathbb{R}^d$ where

$$V := \left\{(t,x,s,y) \in \mathbb{R}^{1+d+1+d} : (t+s, x+y) \in U\right\},$$

and

$$F\big(t,x,(s,y)\big) := f(t+s, x+y), \quad \forall (t,x,s,y) \in V,$$

and the variable (s,y) are considered as parameters.

If $(t_0,x_0) \in U$ denotes an arbitrary point and $B_a(t_0) \times B_b(x_0) \times \big(B_c(0) \times B_c(0)\big)$ is good parametric box. Then, we can apply Theorem 8.1 to prove that $\gamma_{t_0,x_0,s,y}(t) \to \gamma_{t_0,x_0,s,y}(t)$, for every $t \in B_a(t_0)$, as $(s,y) \to (0,0) \in \mathbb{R}^{1+d}$.

The fact that the global domain $D(f)$ is open easily follows from Theorem 8.1, Exercise 8.2 and the following

Exercise 8.3. Let $(t_0,x_0) \in U$ be an arbitrary point and $B_a(t_0) \times B_b(x_0)$ be a good box centered at the point (t_0,x_0) then, given any $a' \in (0,a)$, there is a $\delta > 0$ such that for every point $(\bar{t},\bar{x}) \in B_\delta(t_0) \times B_\delta(x_0)$, there is $b' \in (0,b)$ such that $B_{a'}(\bar{t}) \times B_{b'}(\bar{x}) \subset B_a(t_0) \times B_b(x_0)$ is a good box centered at $(\bar{t},\bar{x})$.

□

9. Exercises

Some important exercises from [VE21, Chapter 2]:
2.2, 2.4, 2.7, 2.10, 2.13, 2.16, 2.17, 2.25, 2.26.

10. Exercises

Some important exercises from [VE21, Chapter 3]:
3.1, 3.2, 3.3, 3.4, 3.7, 3.12, 3.17.

11. Exercises

Some important exercises from [VE21, Chapter 4]:
4.1, 4,3.

12. Exercises

Some important exercises from [VE21, Chapter 5]:
5.1, 5.2, 5.3, 5.4, 5.5, 5.6, 5.14, 5.15, 5.19, 5.21.

13. Exercises

Some important exercises from [VE21, Chapter 6]:
6.1, 6.2, 6.3, 6.4, 6.7, 6.8, 6.11, 6.12, 6.14, 6.17, 6.18, 6.24, 6.25.

14. Exercises

Some important exercises from [VE21, Chapter 7]:
7.1, 7.2, 7.3, 7.5, 7.6, 7.8, 7.9, 7.10, 7.13, 7.18, 7.21.

15. Exercises

Some important exercises from [VE21, Chapter 8]:
8.1, 8.2, 8.3, 8.4, 8.6, 8.9.

References

[Arn06] V. Arnold, *Ordinary differential equations*, Universitext, Springer-Verlag, Berlin, 2006, Translated from the Russian by Roger Cooke, Second printing of the 1992 edition. MR 2242407

[Sot79] J. Sotomayor, *Lições de equações diferenciais ordinárias*, Projeto Euclides, vol. 11, Instituto de Matemática Pura e Aplicada, 1979.

[VE21] M. Viana and J. Espinar, *Differential equations—a dynamical systems approach to theory and practice*, Graduate Studies in Mathematics, vol. 212, American Mathematical Society, Providence, RI, [2021] ©2021, In collaboration with Guilherme T. Goedert and Heber Mesa.

IMPA. Estrada Dona Castorina, 110. Rio de Janeiro, 22460–320, Brazil.
Email address: alejandro@impa.br
URL: w3.impa.br/~alejandro